

GRAHAMSTOWN, South Africa

WMO: 688490

Lat: 33.291S

Lon: 26.503E

Elev: 2106

StdP: 13.61

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.6	41.7	19.0	15.4	65.7	23.2	18.9	62.7	29.3	67.6	26.2	63.3	8.0	290	0.456

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.8	90.3	66.4	86.5	65.8	82.9	65.2	71.3	81.4	69.9	79.1	68.7	77.1	9.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.7	114.0	74.1	67.3	108.7	72.6	66.2	104.4	71.5	36.6	81.5	35.3	79.2	34.2	77.2	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.2	17.8	36.0	99.2	1.5	3.1	34.9	101.4	34.0	103.2	33.1	104.9	32.0	107.1		
(5)			34.0	74.0	1.8	1.3	32.7	75.0	31.6	75.7	30.6	76.5	29.3	77.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	68.4	69.3	66.9	63.7	60.3	56.3	56.0	57.0	57.9	60.6	62.7	65.7		
	DBStd	7.65	5.53	5.76	6.08	6.69	5.91	5.91	6.56	6.57	6.32	6.64	5.89	5.72		
	HDD50	54	0	0	0	1	2	14	19	11	6	1	0	0		
	HDD65	1755	29	19	49	104	169	266	285	263	228	171	112	60		
	CDD50	4445	570	540	524	412	323	201	205	227	245	330	382	486		
	CDD65	667	134	139	107	65	24	4	6	14	16	35	43	80		
	CDH74	4372	784	850	715	371	137	18	37	114	164	326	318	538		
	CDH80	1396	255	306	253	104	17	0	2	21	44	117	90	187		
(14) Wind	WSAvg	8.6	9.1	8.5	7.6	7.6	7.8	8.9	9.2	8.7	8.7	9.3	9.2	9.2		
(15) Precipitation	PrecAvg	24.9	2.1	2.2	2.4	2.2	1.1	1.5	1.4	1.9	2.0	2.8	2.8	2.4		
	PrecMax	34.0	3.9	4.6	5.1	5.5	4.3	5.7	4.8	8.4	6.9	7.4	10.5	9.4		
	PrecMin	15.2	0.9	0.6	0.3	0.4	0.2	0.1	0.0	0.1	0.5	0.6	0.2	0.4		
	PrecStd	5.2	0.7	1.1	1.1	1.4	1.0	1.4	1.3	2.0	1.7	1.5	2.1	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.5	94.1	93.4	89.0	82.2	76.3	77.9	83.2	86.1	90.1	89.8	92.4		
		MCWB	68.3	68.1	66.4	64.3	58.3	53.5	53.0	54.0	57.4	61.2	67.0	66.2		
	2%	DB	86.5	88.1	87.2	83.4	77.8	72.7	74.0	77.0	78.5	83.0	82.7	85.6		
		MCWB	69.0	69.0	65.5	62.9	56.8	52.1	51.9	53.7	57.4	61.8	64.6	66.8		
	5%	DB	81.8	83.2	81.7	78.3	73.7	69.4	70.5	72.2	73.3	76.7	77.4	79.7		
		MCWB	68.0	68.2	65.6	62.4	56.3	51.3	50.9	52.8	56.5	60.7	63.6	65.9		
	10%	DB	77.8	78.9	77.3	73.8	69.6	65.7	66.5	67.5	68.7	71.9	72.8	75.3		
		MCWB	66.7	68.0	65.2	61.8	55.5	50.7	50.3	51.9	55.5	59.3	62.8	64.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.9	72.7	72.1	68.4	63.1	58.2	57.4	60.2	63.4	67.5	71.9	71.1		
		MCDB	84.6	83.3	82.1	77.5	70.7	64.2	66.3	67.8	73.0	79.8	76.9	83.6		
	2%	WB	70.7	71.3	69.7	66.4	60.3	55.7	55.4	58.2	61.0	64.6	67.8	69.2		
		MCDB	81.0	81.1	77.9	74.5	68.8	64.4	64.3	65.6	70.3	74.4	78.2	79.1		
	5%	WB	69.1	70.0	67.9	64.8	58.9	54.1	53.9	56.6	59.1	62.7	65.5	67.6		
		MCDB	78.0	78.6	76.4	72.9	67.5	62.5	62.2	63.5	66.7	71.7	73.8	75.7		
	10%	WB	67.6	68.7	66.4	63.3	57.5	52.7	52.5	54.8	57.5	60.9	63.6	65.9		
		MCDB	75.4	76.4	73.9	70.9	65.9	61.1	60.9	62.3	64.1	68.9	70.2	72.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.0	16.8	16.8	16.5	17.1	17.5	17.7	18.0	17.9	17.5	16.8	16.4		
		MCDBR	24.9	25.5	26.2	23.9	23.3	22.2	22.9	25.8	27.4	27.9	26.1	25.8		
	5% WB	MCWBR	9.0	8.4	8.1	7.8	7.7	8.0	7.9	8.3	9.1	9.9	10.4	9.8		
		MCWBR	21.0	21.0	20.0	19.1	18.5	17.7	17.7	18.1	20.6	21.7	21.8	20.5		
(40) Clear-Sky Solar Irradiance	taub		0.360	0.362	0.347	0.324	0.308	0.294	0.297	0.337	0.359	0.369	0.352	0.356		
		taud	2.491	2.494	2.512	2.551	2.565	2.577	2.563	2.443	2.378	2.385	2.470	2.494		
	Ebn at Noon		310	303	296	286	275	271	276	277	287	297	311	312		
		Edh at Noon	36	35	32	28	24	22	24	30	36	39	37	37		
(44) All-Sky Solar Radiation	RadAvg		2162	1972	1657	1288	1006	899	962	1204	1562	1841	2104	2251		
	RadStd		103	97	122	67	40	47	54	77	106	141	119	145		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.63	+0.76	N/A	N/A	N/A	N/A	N/A	-153	+256	+81	

Nomenclature: See separate page